12th hacker series

Biotechnology and its applications

Introduction

- Biotechnology: Essentially deals with industrial scale, production of bio pharmaceutical and biologicals using microbes, fungi, plants and animals
- Application of biotechnology include:
- 1) therapeutics
- 2) diagnostics
- 3) genetically modified crops for agriculture
- 4) processed food
- 5) bioremediation
- 6) waste treatment
- 7) energy production
- Three critical research areas of biotechnology:
- 1) providing best catalyst in the form of improved microbes or pure enzymes
- 2) creating optimal conditions through genetic engineering
- 3) downstream processing technologies for purification

Biotechnological application in agriculture

- Food production could possibly be increased in three ways:
- 1) agrochemical based agriculture
- 2) organic agriculture
- 3) genetically engineered crop-based agriculture
- Green revolution resulted in tripling a food production
- Reasons for success of green revolution:
- 1) improved crop varieties
- 2) agrochemicals (fertilisers + pesticides)
- 3) better management practices

- problem area and hindrance:
- 1) enhancement in food production by green Revolution was still not enough to feed a growing population
- 2) Agro chemicals are often too expensive for farmers of developing world
- 3) increase in yield with existing varieties is not possible using conventional breeding

Genetically modified organisms

- organisms, including plants, bacteria, fungi, and animals, whose genes have been altered by manipulation are called genetically modified organisms. (GMO.)
- Applications of genetic modification:
- 1) made crops more tolerant to abiotic stress (cold, drought, salt, heat)
- 2) reduce reliance on chemical pesticides (pest resistant crops)
- 3) helped to reduce post-harvest losses
- 4) increase efficiency of mineral usage by plants (prevents early exhaustion of fertility of soil)
- 5) enhance nutritional value of food, eg, golden rice i.e, vitamin A enriched rice
- Tailor made plants- plants have been developed to supply alternative resources to industries in form of star fuels and pharmaceuticals

Insect resistant plants

- provides resistance to insects without the need of insecticides
- Examples of biopesticides are Bt corn, Bt cotton, rice, tomato, potato, and soyabean
- bacillus thuringenesis (A bacterium)

Source of cry gene

↓

codes for Bt toxin → Inactive protoxins (proteins) (crystals)

produced during a particular phase of their growth

Endotoxin

Sequence of events

Cry genes from B. thuringiensis

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Bt cotton (cells express inactive protoxins)

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Ingested by cotton bollworm

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Alkaline pH in the midgut of insects solubilizes protein crystals

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Activated toxins

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create pores in midgut epithelial cells

↓

swelling of midgut epithelial cells & thereby lysis

↓

Death of insect

- choice of gene depends on:
- 1) target pest
- *cryIAc* and *cryIIAb* - cotton bollworm
- *cryIAb* - corn borer
- Bt toxins are insect group specific :
- Lepidoptera - tobacco budworm, armyworm, cotton, bollworm, corn borer
- Coleoptera - beetles
- Diptera- flies and mosquito

Pest resistant plants

- method of cellular defence seen in all eukaryotes against pest infestation
- Technique responsible: RNA interference
- Based on post-transcriptional silencing of mRNA
- Translation of mRNA coded from pest-specific genes is silenced / prevented due to formation of complementary dsRNA

↓ source

↓ Viruses with RNA genome

↓ Mobile genetic elements Transposons replicating via an RNA intermediate

- the cause of nematode resistant transgenic tobacco:
- Passed, causing roots knot disease in tobacco plant: meloidogyne incognita (nematode/ helminth)
- Nematode specific gene is introduced in host plant by using Ti plasmid of Agrobacterium tumefaciens, in such a manner that it produces both sense and antisense RNA in host cells

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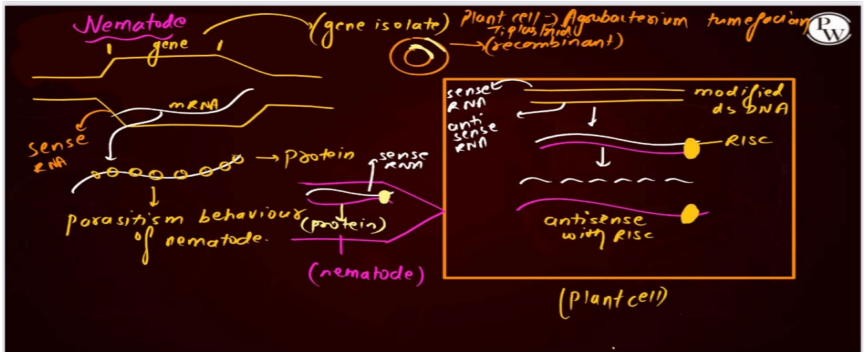
sense RNA and antisense RNA being complementary form dsRNA that initiates RNAi

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parasite could not survive in a transgenic host expressing specific interfering RNA

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Host plant - generated dsRNA triggers protection against nematode infestation

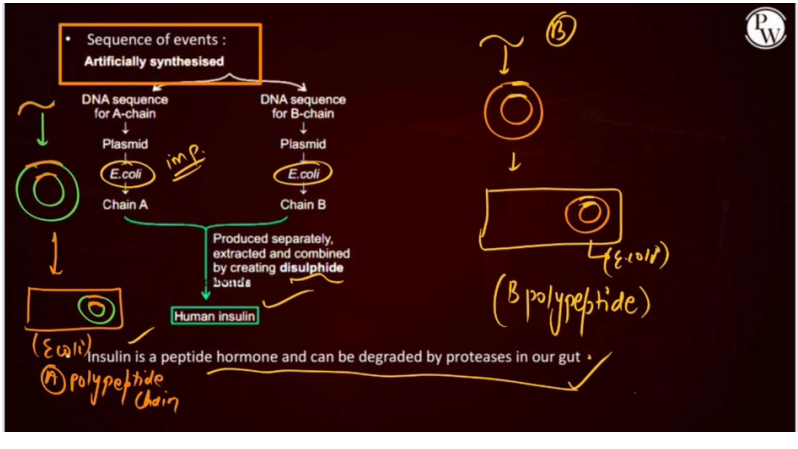


Biotechnological application in medicine

- advantages of recombinant therapeutics: 30 recombinant therapeutics have been approved for human use world over. In India, 12 of these are presently being marketed.
- Mass production of safe and effective drugs
- do not induce unwanted immunological responses

Genetically engineered human insulin

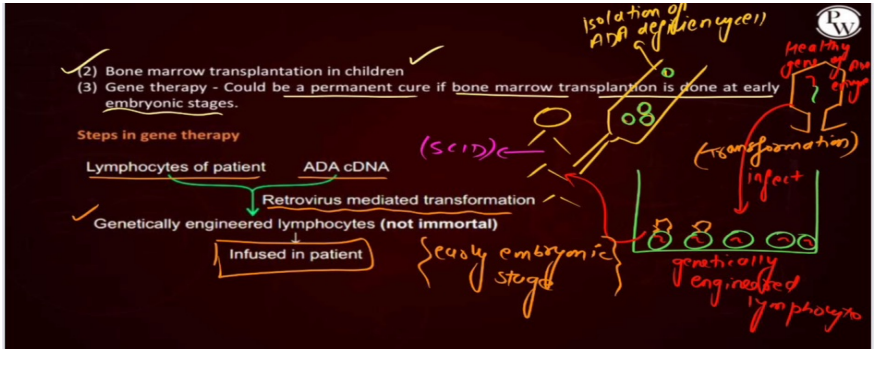
- problem- insulin extracted from slaughtered cattle and pigs could cause allergy
- Solution- production of humulin
- Recombinant insulin manufactured by Eli, Lilly and American company in 1983
- The main challenge of production of insulin using rDNA technique was getting insulin assembled into a mature formed



Gene therapy

- insertion of gene into an individual cell to treat diseases
- 1) replacing a defective mutant allele with a functional one
- 2) gene targeting, which involves gene amplification
- First clinical gene therapy was conducted in 1990 in a four year old girl to treat adenosine deaminase deficiency. ADA enzyme is crucial for immune system to function.
- Treatment for ADA deficiency:
- 1) enzyme replacement therapy- functional ADA is given by injection, not completely curative
- 2) bone marrow transplantation in children
- 3) gene therapy- *Introducing the ADA gene from marrow cells into early embryonic cells may provide a permanent cure.*

Stages in gene therapy



Molecular diagnosis method

parameters → conventional
easy detection
examples
Not possible
serum & urine
analysis

modern
possible
RDT, PCR, ELISA

- PCR(polymerase chain reaction)
- Basis- nucleic acid amplification
- Advantage- detection of very low concentration of bacteria or virus
- Uses- detection of HIV infection, mutation in genes in cancer patients, genetic disorders
- ELISA(enzyme linked immunosorbent assay) :
- Basis- antigen antibody interaction
- Uses- detect the presence of antigen or antibodies synthesised against pathogens
- Autoradiography:
- Probe: radioactive ssDNA or ssRNA that hybridised with complementary DNA
- Probe will not hybridise with mutated gene, hence mutated gene will not appear on the photographic film due to lack of complementarity

Transgenic animals

- possesses manipulated DNA and express foreign gene
- Transgenic, rats, rabbit, rabbits, pigs, sheep, cows, and fish have been produced
- 95% of transgenic animals are mice
- Uses of transgenic animals
- to study how genes are regulated and how they affect the normal functions of the body
- Transgenic models exist for study of diseases like cancer, cystic fibrosis, rheumatoid arthritis, Alzheimer's
- Biological products
- d-l Antitrypsin to treat emphysema
- Similar attempts are made for treatment of PKU (phenylketonuria) and cystic fibrosis
- First transgenic cow: Rosie developed in 1997 produced human protein enriched milk
- The milk contain alpha-Lactalbumin: more balance product for human babies than natural cow milk

Vaccine safety

- transgenic mice are being used to test the safety of polio vaccine to replace the use of monkeys
- Chemical safety testing
- Transgenic animals are made more sensitive to toxic substances to obtain results in less time

Ethical issues

- genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem
- Genetic manipulation of living organism by humans has to regulated for moral and biological significance
- GEAC (genetic engineering approval committee): make decision regarding the validity of genetic modification, research, and the safety of introducing GMO for public services
- Developing countries are rich in biodiversity and traditional knowledge related to bioresources
- Biopiracy: refers to use of bio resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment

Controversies regarding patents and biopiracy

- basmati rice: 2,00,000 varieties of rice in India. 27 documented varieties of Basmati rice in India.
- in 1997, an American Company *patent* and rights on Basmati rice through the US patent and trademark office

Basmati rice × Semi-dwarf variety of rice

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New variety of Basmati rice

- turmeric
- Neem
- the Indian Parliament has recently cleared the second amendment of the Indian patent bill